

# Category Theory and Functional Programming

## Universality and Adjoints

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# Preliminaries

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## Textbook

Abramsky and Tzevelekos (2011). Introduction to Categories and Categorical Logic.

## Convention

The numbers and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook.

# Outline

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## Universality and Adjoints

- Adjunctions for Pre-orders

- Universal Arrows

- Exponentials

- Cartesian Closed Categories

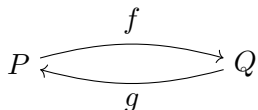
- References

# Adjunctions for Pre-orders

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## Definition

Let  $(P, \preceq_P)$  and  $(Q, \preceq_Q)$  be two pre-ordered sets and let  $f$  and  $g$  be two homomorphisms (monotone maps) forth and back.



The maps  $f$  and  $g$  are **adjoint**, denoted  $f \dashv g$ , when for all  $x \in P$ ,  $y \in Q$ ,

$$f x \preceq_Q y \quad \text{iff} \quad x \preceq_P g y.$$

The map  $f$  is the **left adjoint** and the map  $g$  is the **right adjoint** (Awodey and Bauer 2022).

# Universal Arrows

# Universal Arrows

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## Definition

Let  $G : \mathcal{D} \rightarrow \mathcal{C}$  be a functor and let  $C$  be an object of  $\mathcal{C}$ . A **universal arrow** from  $C$  to  $G$  is a pair  $(D, \eta)$  where  $D$  is an object of  $\mathcal{D}$  and  $\eta$  is an arrow

$$\eta : C \rightarrow G_0 D$$

# Exponentials

# Exponentials

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## Introduction

We shall introduce abstract characterisations of the following facts of set theory:

- (i) The functions of a **set**  $X$  to a **set**  $Y$ , denoted  $Y^X$ , are also a **set**.
- (ii) A function of **two** variables  $f : X \times Y \rightarrow Z$  can be represented as a function of **one** variable  $\lambda f : X \rightarrow Z^Y$ .



# Exponentials

## Set theory

Let  $Z^Y$  be the set of functions from the set  $Y$  to the set  $Z$ . For any set  $X$  and function  $g : X \times Y \rightarrow Z$ , the following diagrams commutes:

$$\begin{array}{ccc} & Z^Y & \\ \uparrow \Lambda g & & \\ X & & \end{array} \quad \begin{array}{ccc} Z^Y \times Y & \xrightarrow{\text{eval}_{Y,Z}} & Z \\ \uparrow \Lambda g \times \text{id}_Y & \nearrow g & \\ X \times Y & & \end{array}$$

$$(g = \text{eval}_{Y,Z} \circ (\Lambda g \times \text{id}_Y))$$

where

$$\text{eval}_{Y,Z} : (Z^Y \times Y) \rightarrow Z := (f, y) \mapsto f y$$

(application),

$$\Lambda g : X \rightarrow (Y \rightarrow Z) := x y \mapsto g(x, y)$$

(currying).

# Exponentials

## Definition

Let  $\mathcal{C}$  be a category with binary products, and let  $Z$  and  $Y$  be objects of  $\mathcal{C}$ . The pair  $(Z^Y, \text{eval})$ , where  $Z^Y$  is an object of  $\mathcal{C}$  and  $\text{eval}$  is an arrow

$$\text{eval} : (Z^Y \times Y) \rightarrow Z$$

is an **exponential object** iff for any object  $X$  and arrow  $g : X \times Y \rightarrow Z$ , there is a **unique** arrow  $\Lambda g : X \rightarrow Z^Y$  such that the following diagram commutes:

The diagram illustrates the universal property of the exponential object  $Z^Y$ . It consists of the following components:

- Objects:  $X$ ,  $X \times Y$ ,  $Z^Y$ ,  $Z^Y \times Y$ , and  $Z$ .
- Arrows:
  - A vertical dotted arrow from  $X$  to  $Z^Y$  labeled  $\Lambda g$ .
  - A vertical dotted arrow from  $X \times Y$  to  $Z^Y \times Y$  labeled  $\Lambda g \times \text{id}_Y$ .
  - A diagonal solid arrow from  $X \times Y$  to  $Z$  labeled  $g$ .
  - A horizontal solid arrow from  $Z^Y \times Y$  to  $Z$  labeled  $\text{eval}_{Y,Z}$ .

The diagram commutes, meaning  $g = \text{eval}_{Y,Z} \circ (\Lambda g \times \text{id}_Y)$ .

$$(g = \text{eval}_{Y,Z} \circ (\Lambda g \times \text{id}_Y))$$

# Cartesian Closed Categories

# Cartesian Closed Categories

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## Definition

A category is a **Cartesian closed category** iff it has finite products (i.e. a terminal object and binary products) and exponentials.

## References

# References

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S. Abramsky and N. Tzevelekos (2011). Introduction to Categories and Categorical Logic. In: New Structures for Physics. Ed. by Bob Coecke. Vol. 813. Lecture Notes in Physics. Springer, pp. 3–94. DOI: [10.1007/978-3-642-12821-9\\_1](https://doi.org/10.1007/978-3-642-12821-9_1) (cit. on p. 2).



Steve Awodey and Andrej Bauer (2022). Introduction to Categorical Logic. Appendix A. Category Theory. Draft version 2022-01-15. URL: <https://awodey.github.io/catlog/> (cit. on p. 4).